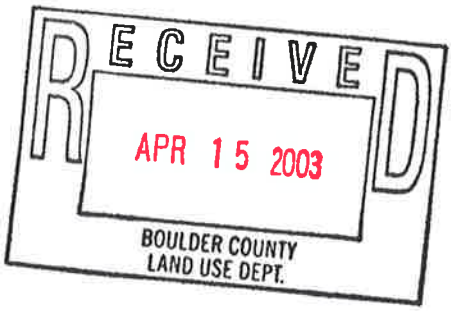
CEMEX

April 14, 2003



Mr. Graham Billingsley
Director
Boulder County Land Use Department
Courthouse Annex Building
2045 13th Street
Boulder, CO 80306

RE: Lyons Cement Plant—Tire Derived Fuel Use
Request for a Determination Concerning Use by Special Review #SU-88-21

Dear Mr. Billingsley,

The following information is provided as an addition to the recent letter from Banks and Gesso, LLC, dated April 10, 2003, for consideration by the Boulder Board of County Commissioners (BOCC) in support of our position that the above referenced special use permit has not lapsed.

Activities to repair and improve the tire chute at the Lyons Cement Plant in preparation for future use of Tire Derived Fuel (TDF) include the following (please see attached contractor notes and invoices):

May 1995—Refractory contractor completed work to reactivate the tire chute on the kiln feed hood.

1996 Kiln Outage—Tire chute was inspected and measured to establish parameters for future use.

1997 Kiln Outage—Removed damaged tire chute and patched with new metal and refractory.

2000 Kiln Outage—New tire chute installed on kiln feed hood.

Current project milestones related to TDF use include:

Summer of 2001—I met with Rick Grice and Olga Erlich of the State of Colorado Governor's Office of Energy to discuss the State of Colorado's position on using tires as fuel. I was informed that the State of Colorado supported such efforts and

United States Operations
P.O. Box 529, Lyons, Colorado. 80540. USA. Phone: (303) 823-2100, Fax: (303) 823-2199

that economic incentives were available to support such private industry initiatives.

February 2002—Lyons Plant hosted a meeting of technical resources to determine the feasibility and most successful approach to using tires as fuel in the Lyons kiln. As a result of this meeting, an injection point and mechanism was identified for a future test.

June 2002—Cemex announced the company's intent to complete a test of Tire Derived Fuels to the Northern Boulder County Environmental Health Task Force. This was an organization made up of representatives of local companies, community, and environmental regulatory bodies.

October 2002—Cemex placed ads in the local newspapers announcing the TDF test which would include a greatly extended suite of elements and compounds. It also announced an open house to be held to answer questions from the community at the Lyons elementary school.

November 2002—Cemex completed the test of TDF on the Lyons kiln.

January 2003—Laboratory test results available for evaluation by a professional health expert.

May 2003—Cemex will announce the test results and again invite the community to an open house to answer community questions about TDF use.

We appreciate your consideration of the information submitted in this request.

Sincerely,

John Lohr
Plant Manager, Lyons Cement Plant

attachments

cc: Mr. Ray Petros, Esq., Petros & White, LLC
Mr. Paul Banks, Banks and Gesso, LLC


IMPACT
B
Suspect avoids execution
ICE TIME

1995
OUTAGE

Southdown Lockout Policy
Lyons Plant

We are committed to a safe operation at our facility. During repair work, we typically must work on equipment that could injure of kill an individual if it were operated or energized during the work. We have developed a lock out policy with the intent of minimizing this risk. All employees, managers, contractors, and contract employees are required to abide by this policy. We will insist on 100% compliance.

Contractor Policy
Follow all aspects of the written lock out policy at all times with all personnel.

Failure to install or remove a lock according the policy will be dealt with as follows:

1. The first incident for the contractor and for the contract employee involved will result in a warning.

2. The second incident involving a contract employee the company will request that he be removed from the job site.

3. The second and subsequent violations from a contractor will result in a fine being levied of $100 per violation.

Southdown, Inc.

John Lohr
Plant Manager

Contractor: WESTERN REFRACTORY CONSTRUCTION, INC.

Name: Nadine Woodside

Title: Sec-Treas.

Date Reviewed: 5-08-95

CONTRACTOR'S AGREEMENT

THIS AGREEMENT is made as of this ___ day of _____, 19__, by and between Southdown, Inc., a Louisiana corporation with an office at 5134 Ute Highway, Lyons, CO __ (the "Company"), and Western Refractory, Inc. a Colorado corporation with an office at 4023 Mulligan Drive, Longmont, CO (the "Contractor").

RECITAL

The Company desires to engage the Contractor to perform certain construction work for the Company, and the Contractor desires to accept such engagement, upon and subject to the terms and conditions of this Agreement.

NOW, THEREFORE, in consideration of the mutual covenants contained herein, the Company and the Contractor hereby agree as follows:

ARTICLE 1
ENGAGEMENT AND SCOPE

1.1 The Company hereby engages the Contractor to provide the aforementioned construction work in accordance with the terms of this Agreement and the Purchase Order, dated _____, 199_ (attached as Exhibit "A") at the Company's facility located at Lyons, CO (the "Facility"), with respect to Refractory Tearout and Replacement (the "Work"), and the Contractor accepts such engagement.

1.2 The Contractor shall provide and perform all necessary supervision, labor and services and provide and furnish all necessary supplies, materials, tools and equipment required to complete the Work at the Facility in accordance with this Agreement.

1.3 The Work shall be performed by the Contractor as an independent contractor and under the sole supervision, management, direction and control of Contractor in accordance with the terms and conditions of this Agreement.

1.4 The Contractor shall be responsible to the Company for the acts and omissions of its employees, Subcontractors and other persons performing any of the Work for the Contractor. The Contractor shall employ on the work the proper number of experienced, skilled personnel necessary to perform the Work and shall at all times enforce strict discipline and good order among its employees and Subcontractors. The Contractor shall require its employees and Subcontractors to comply with the general standards and rules of the Facility.

ARTICLE 2
TERM

2.1 The Contractor shall commence the Work on May 5, 1995, and shall complete the same on or before May 15, 1995, (the "Term").

2.2 In the event Contractor fails to commence said Work within the time specified, or having begun said Work abandons it for any reason, suspends or refuses to continue it, or defaults in any manner in the performance under the terms of this Agreement for a period of five (5) days (unless Contractor is prevented from continuing for reasons beyond its control), the Company shall have the right to take over said Work and complete it or have said Work completed by another in any reasonable manner at Contractor's expense.

ARTICLE 3
COMPENSATION AND METHOD OF PAYMENT

3.1 Company shall pay Contractor a total fee of 28,293.00 Dollars ($28,293.) for completion of the "Bid" work in addition to the "Time and Materials" Work . The Time and Materials work will be billed in accordance with the attached list of charges for all work requested by Company.

3.2 Contractor shall submit a single invoice to the Company for the entire Bid and Time and Materials work upon completion of the Work.

3.3 Company shall pay the full amount of the invoice within forty-five (45) days of its receipt. If, however, Company objects to all or any portion of the invoice, Company shall so notify Contractor of the same within fifteen (15) days from date of receipt of the invoice, stating the reasons for the objection. Company shall be entitled to withhold payment of any amounts in dispute. The parties shall immediately make every effort to settle the disputed portion of the invoice.

ARTICLE 4
INDEMNITIES AND INSURANCE

4.1 Both the Company and the Contractor agree to indemnify and hold the other party harmless from any and all losses, damages, costs, penalties, expenses (including, but not limited to, reasonable investigation or legal expenses), liabilities, judgments, liens, suits, claims or demands arising out of any actual or threatened damage to property, including the other Party's property, or injuries to or death of persons, including the other Party's employees, to the extent caused by or resulting from the indemnifying Party's (or its Subcontractors or any other person performing any of the Work for the indemnifying Party) acts or omissions at the
Facility or in connection with the Work, or any breach or threatened breach of this Agreement by the indemnifying Party, excluding and to the extent of, the other Party's negligence or willful misconduct.

4.2 The Contractor agrees at all times during the Term of this Agreement to maintain in full force and effect insurance in accordance with Article 4.3. Before commencement of any of the Work hereunder, the Contractor agrees to furnish to the Company certificates of insurance or other evidence satisfactory to the Company to the effect that such insurance has been procured and is in force.

4.3 For the purpose of this Agreement, Contractor shall carry the following types of insurance in at least the limits (which may be a combination of primary and excess coverage) specified below:

COVERAGES	LIMITS OF LIABILITY
Worker's Compensation	Statutory
Employer's Liability	$500,000
Comprehensive General Liability (Incl. bodily injury and property damage)	$1,000,000 per occurrence
Automobile Liability (Incl. bodily injury and property damage)	$1,000,000 per occurrence

ARTICLE 5

SUBCONTRACTS

5.1 A Subcontractor is a person or entity who performs any part of the Work and who is not a direct, fulltime employee of Contractor. The Contractor shall be responsible for any portion of the Work performed by a Subcontractor.

ARTICLE 6
PROTECTION OF PERSONS AND PROPERTY

6.1 Contractor acknowledges and is aware that the Facility is the site of heavy industrial activity which may involve the presence and/or use of hazardous substances and/or industrial wastes, and Contractor knowingly and voluntarily assumes all risk of injury and damage to Contractor and Contractor's property, employees, Subcontractors and others working for the Contractor, caused by exposure to such hazardous substances and/or industrial wastes. Contractor agrees to advise fully all of its employees, Subcontractors and others working for the Contractor at the Facility, of the Facility risks and of all necessary environmental, safety and health procedures required by the Company. Contractor warrants that any employee or Subcontractor that will be working at the Facility and subject to such potential exposure has received, prior to performing any Work, all training required by applicable federal, state and local laws, rules and regulations, including, but not limited to, the Mine Safety and Health Act of 1977, as amended, and Title III of the Superfund Amendments and Reauthorization Act.

6.2 In compliance with the requirements of the Mine Safety and Health Act of 1977, as amended, the Contractor shall supply the Company with an Independent Contractor Register including the following information prior to commencing the Work: 1) the Contractor's trade name, business address and business telephone number, 2) a short description of the nature of the Work to be performed by the Contractor and where the Work is to be performed, 3) the Contractor's Mine Safety and Health Administration identification number, if any; and 4) the Contractor's address for record of service of citations or other documents involving the Contractor.

ARTICLE 7
TRADE SECRETS AND CONFIDENTIAL INFORMATION

7.1 Contractor agrees that it will not at any time, either while engaged hereunder by Company or afterwards, make any outside use of, or disclose to any other person or organization, except as authorized in writing by Company, or as compelled by lawful order, any information, whether patentable or not, regarding plans, programs, facilities, designs, processes, products, costs, equipment, operations or customers of Company which comes within the knowledge of Contractor in the performance of the Work hereunder.

7.2 Contractor recognizes and acknowledges that it may have access to certain confidential information of the Company and that such information constitutes valuable, special and unique property of the Company. Contractor will not, during or after the term of this Agreement, use or disclose any of such confidential information to any person, firm, corporation, association or other entity, except to authorized representatives of the Company for any reason or purpose whatsoever, other than in furtherance of this Agreement. In the event of a breach or threatened breach by Contractor of the provisions of this Article 7, the Company, shall be entitled to an injunction restraining Contractor from using or disclosing, in whole or in part, such confidential information. Nothing herein shall be construed as prohibiting the Company, from pursuing any other remedies available to them for such breach or threatened breach, including the recovery of damages from Contractor.

7.3 The obligations of this Article 7 do not apply to information which: (a) is or becomes part of the public domain without the breach of any obligation of confidentiality owed to the Company; (b) is lawfully in the possession of Contractor at the time it was acquired hereunder without the breach of any obligation of confidentiality owed to the Company; or (c) is required under law to be disclosed in public.

ARTICLE 8
MISCELLANEOUS

8.1 The Company may, without prejudice to any other right or remedy, upon giving the Contractor thirty (30) days written notice, or immediately for any breach of this Agreement, terminate this Agreement. The terms of Articles 4 and 7 shall survive the expiration or termination of this Agreement.

8.2 This Agreement shall be subject to and governed by the laws of the State in which the Facility is located. The Work and performance of same shall comply with all city, county, state and federal codes, rules, regulations and orders.

8.3 Failure to insist upon strict compliance with any provision hereof shall not be deemed a waiver of such provision or any other provision hereof.

8.4 This Agreement, with its exhibit, constitutes the entire agreement and understanding between the Parties and supersedes an prior agreement and understanding, whether written or oral, relating to the Work.

8.5 This Agreement may be executed in one or more counterparts, each of which shall be deemed to be an original, and such counterparts shall together constitute and be one and the same agreement.

IN WITNESS WHEREOF, the parties have executed this Agreement effective as of the day and year first above written.

SOUTHDOWN, INC. (Company)

By: _____

Name: John Lohr Andrew Jackura

Title: Plant Manager Director of Maintenance

Western Refractory, Inc. (Contractor)

By: _____

Name: _____

Title: _____

Please provide labor and equipment necessary to complete the following projects:

TERTIARY AIR DUCT REPAIR

Description - The refractory has failed in a 20 foot section along the top of the tertiary air duct (8 foot diameter duct). Subsequently the metal in the same area also burned up. A refractory patch has been poured over the top of the hole. The inside of the duct is approximately half full of clinkered material, some or all of which will have to be cleaned out to accommodate your work.

1. Remove refractory from the top half of the duct. The estimated size of the area to be repaired is 12' X 20'.

2. Remove the damaged metal, along with the patch material from the top of the duct. Southdown will furnish replacement metal in 4'X 8'X 1/4" sheets rolled to the proper dimension. The length of each piece will be 4' along the duct with a circumferential length of 8'.

3. Remove castable and material from the inside of the duct. A chute will have to be constructed to dump the material from the duct to the ground level in such a way as to limit dust. There will be other crews working in the same general area (burner floor, cooler etc.). The distance from the center of the duct to the ground is 42 feet. Material must then be moved to an area outside the cooler room where S.W.P.C. will haul it away.

4. Remove anchors from any remaining exposed metal. All old metal must be ground flush and sandblasted before new anchors are installed.

5. Place and weld new metal as needed.

6. Install 5" 304 S.S. anchors on 9" centers. Each anchor will be staggered and rotated 45 degrees. Anchors will be supplied by S.W.P.C.

7. Install 2 1/2" of lightweight insulating material.

8. Form with 3/4" smooth sided plywood treated with form release prior to installation. Stiffeners will be of sufficient size and number to give a well supported and smooth pour. Form will be installed to allow a 6" thickness of castable, (8 1/2" inches total thickness including lightweight).

9. Install castable. Lightweight and castable will be supplied by S.W.P.C.

10. Remove forms after allotted cure time.

Bid Price = $28,923.00

I.K.N. COOLER AND CURBS

1. Remove refractory curbs that cover the stainless steel plates over the spare I.K.N. grates. Exposed grates must be covered with plastic and 3/4" plywood for protection. All old anchors on the cover plates will be removed and ground flush. Refractory must be removed from the cooler and piled outside the cooler room where it will be hauled away by S.W.P.C..

2. After S.W.P.C. personnel have completed necessary cooler repairs Western will install new anchors on the cover plates. Anchors (supplied by S.W.P.C.) are 8" 310 S.S. Anchors will be welded on 18" centers with anchors being staggered and rotated 45 degrees. Arrangement must be made to attach the new pour to the existing castable.

3. Forms will be smooth sided 3/4" plywood treated with form release prior to installation. Stiffeners will be of sufficient size and number to insure a well supported smooth pour.

4. Castable will be supplied by S.W.P.C. Castable may be poured or pumped.

5. Forms will be removed following allotted cure time.

T & M

MISCELLANEOUS LABOR REQUIREMENTS

1. Provide one brick mason and one helper around the clock to assist S.W.P.C. in replacing kiln brick.

T & M

POSSIBLE TIME AND MATERIAL WORK

1. Remove castable plug from the tire chute at the second level of the preheater and repair any castable damage.

2. Repair any damage to the expansion joint in the tertiary air duct north of the drop out chamber.

3. Provide necessary repairs to steel and refractory at multiple areas in and around the clinker cooler and kiln hood.

4. Repair any refractory damage at the kiln tail ring and

1995

Tire Chute

The chute for injecting whole tires to the kiln feed hood was opened up.

The existing castable was removed by jackhammer. New anchors were welded on and the tie-in was gunned with Tuff-Gun 28. This area will need to be watched closely for buildup and refractory failure.

1995 Outage

Preheater

1. The north wall below the floor of level 3 was bad. This castable was removed and the steel was replaced. 2 new 9" poke ports were installed. This area was poured with B.M.I. C-60 HS.

2. Sometime after the initial inspection of the tower some brick fell out of the south side of the precleaning cyclone. This area was formed and gunned from the outside with Tuff-Gun 28. This was a 2'x2' patch. The remained of this wall is origional brick which is very loose and pulled away from the lightweight insulation about 1½". The lightweight is also pulled away from the wall about 2". Expect to have problems in this area before next scheduled shutdown.

A decision needs to be made soon on the type of repairs that are going to be made in this area. This is going to be a major project involving considerable time and expense.

3. The wall extension on the shelf between the combution chamber and the precleaned is starting to fail. This will need replaced next year. We need to decide if this wall is really needed and if it is the proper size.

4. The vertical bullnose on the west side of the combustion chamber exit duct was repaired with Tuff-Gun 28. This entire corner should be rebuilt next shutdown.

5. The bottom section of the Tertiary air damper was replaced. The bullnose on top of the duct on the north side of the damper was torn out and replaced with new steel and castable.

6. A small section (2'x3') of castable was replaced on the east side of the burl chamber. This area is in good shape and we should experience no problems there.

7. On the west wall of level 3½ some damage to the castable was found. This was a very small area. (1'x1') This damage appears to have been caused by careless jackhammer use. No repair was made to this area because I felt the remaining castable is still better than a patch would be.

8. The fire chute was opened up. Anchors were installed and the tie-in was made with Tuff-Gun 28. This area will need to be watched for buildup and castable failure.

JOB PROGRESS REPORT

CUSTOMER Southwestern Portland Job # All

ADDRESS Provo CO Date 5-12-95

Report # 5 DAYS

JOB PROGRESS 1050 95 — Patched hole in wall welded anchors & shot Light waight 2" Thick Above Dryer Removed old Refractory Ready for RAM

10495 Cooler cleaned anchors after Shooting Lightwaight. Stocked Lumber For Forms

10595 T-Duct Trm Formed & got Ready to Pour on Sat. We worked Late To Finish This Form. night Shift started on Cooler

DELAYS CAUSED BY OTHERS 10495 — worked on chipping out fire Shut

COMMENTS

WESTERN REFRACTORY CONSTRUCTION, INC.

Superintendent

JOB PROGRESS REPORT

CUSTOMER South western Portland Job # 211

ADDRESS Lyons Col Date 5-13-95

Report # 6 Days

JOB PROGRESS 103795 - Pumped Refractory in Teriary Duct 104595. 4" Pumped Refractory in Rest of Teriary Duct Except up Take OFF Cooler.

104995 Pumped Cooler Side walls.

104895 Chipped on Fire Shut were having a Hard Time getting it out its about 16" TO 18" Thick.

105095 - Dryer started Forming Walls at Shift End it was about 75% Formed.

DELAYS CAUSED BY OTHERS

COMMENTS

WESTERN REFRACTORY CONSTRUCTION, INC.

Superintendent

JOB PROGRESS REPORT

CUSTOMER SWP Job #

ADDRESS Lyons CO Date 5/14/95 Sun

Report # 7 Nights

JOB PROGRESS 104995 Cooler - Finished Forming west and North Curb and poured Castable. Weld Anchors on (Patch) Roof Area up From Clinker Breaker and Installed Refractory. Cleaned up work Area Mixer Etc. Bull Nose

103795 Tertiary Duct - [redacted] 104595 Tertiary Duct T+M - Removed Forms and had a good Start on Lowering to Ground

104895 Preheat Tower - Continued Tear-out of The Tire Shoot.

DELAYS CAUSED BY OTHERS

COMMENTS So Far The pours Look Good!

WESTERN REFRACTORY CONSTRUCTION, INC.

Wayne A Curtis

Superintendent

JOB PROGRESS REPORT

CUSTOMER South western Portland Job # 811

ADDRESS Lyons Co Date 5-14-95

Report # 7 thru 15

JOB PROGRESS 10795 Contued work - We Removed Riggins + weld Pour Holes closed

10495 T Duct T+M Removed Riggins + weld Pour Holes Finished welding out new Steel. Built ~~Door~~ man way Door on Target Drop out.

10495 STILL Trying to get Tire ~~Shot~~ Shut opened up.

10495 Cooler Forming Curbs + Poured East Side

DELAYS CAUSED BY OTHERS 10595 Dryer

Poured Target Wall + Around door Rammed + Gunned up ~~Part~~ of the Dryer

Bob Caplet

COMMENTS Mark Time Sheet for Burner P.M. Removed and Bonded Brick (2) for Dryer

WESTERN REFRACTORY CONSTRUCTION, INC.

Superintendent Bob Caplet

JOB PROGRESS REPORT

CUSTOMER Southwestern Portland Job # 911

ADDRESS Lyons Co Date 5-15-95

Report # 8 17 95

JOB PROGRESS We are working IN All AreAS Clean up Stripping Form Have Its Safe We Fix A Bad Spot on 6th Leval Shoot Bullnose Patch on 6th Leval Finished Chipping out Fire Shut Welded Anchors + Shot Gunite To Square up The Corner. Shot Bad Spots in By Pass orfice. Packed Expantion Joint with Ko Wool in Riser Line Removed Scaffolding in Riser + cleaned out old Refractory Rubble in Swirl Chamber

DELAYS CAUSED BY OTHERS Patched around Door in Tertiary Dropout Poured new Door Plug

COMMENTS

WESTERN REFRACTORY CONSTRUCTION, INC.

Superintendent

For SWPC

SHUTDOWN LIST FOR LONG AND SHORT OUTAGES 1996

1) Weld around the SE door level 6 on the PC Cyclone.
2) Bottom side of the Alk fan duct to the stack, crack.
3) Improve lighting in the under grate comp.
4) S side or belly doors on the Pug Mill
5) Open E side Pug level of the dust silo.
6) 9" ports on the tower.
7) 78 screw belly doors, straighten/ stiffen.
8) 78 screw two gate feeders
9) Weld around the kiln camera door
10) Replace Sankyo diaphram.
11) Install 9" ports E&W sides of cooler: chain pull
12) Hole in Kiln duct 20'S of ST high E side.
13) Large crack Alk duct 4' S of tower.
14) Clean/replace ST manifold.
15) Dryer CC Gnd level door frame cracked.
16) Rebuild outside clinker discharge (29 belt discharge)
17) 529-22 2nd return roller from the south
18) 529-22 south side east end tail pulley wearing through
19) Blu ram 2" pipe in north hole east side of hood
20) 3rd level east side 9'' ports need welding
21) 3rd level east side replace 4' with new metal and castable
22) EAST and West Kiln feed hood door frames need rebuilt
23) 6th level west side 6' hot spot needs new refractory
24) Install rubber tips on bleed in air dampers
25) Repair kiln roll back brake
26) Check tower coal burner "A" to see if it is plugged.
27) Kiln discharge seal rubbing
28) Oil leak on the RM discharge end
29) Kiln feed end seal needs repair
30) 78 screw needs reworked
31) Clean out screw discharge of kiln (529-93) needs repair
32) Rebuild chutes with new gates going to 21 and 22 drag
33) Rebuild drag chains in pit and cooler
34) Install load bed under primary crusher
35) Rebuild primary feed hopper,it is about to fall down
36) Rebuild or repair reclaim system end to end
37) Rebuild raw mill transport line to allow for more tonnage
38) Replace the cooler camera with a color camera
39) Move or hang the portable kiln shell fans so they are off the walk
40) Repair lights west wall "A"
41) Metal at 229-1 feed point to prevent side spills
42) Add a air cannon E side just below the floor level
43) Gas valve leaking level 3 of tower W/O 047250
44) Inspect air cannons W/O 047254
45) Pull chains on comp. 5 and 7 Alk side
46) new glass for the baghouse inspection doors.
47) Measure the inside of the tire chute

1997 OUTAGE

Rich: For 4-1-97

1) Kiln Tail Ring: pour or (gun?)
   Anchors 3045s 8" OR 7" 12" CENTERS
   Material what/how much ? 10,497 LBS + REBOUND
   Wood for forms?
   How will this fit into the schedule

2) Tire Chute: Steel 3/8 or (1/2?) BUTT WELDED
   Material what/how much VERSAFLOW 58 12 CFT-NO
   9" Fuller ports - 2
   Anchors 3045S 8"
   ok

3) Kiln Nose: Plywood, 2x12, steel forms DONE

4) Cooler Camera: Concrete Coring-when? BUM ok
   JLG
   2 1/2" SCH 80 310SS pipe FOUND

5) Locate caps for poke ports on level 6 DONE AND ON

6) Move 310SS half rings from south of SR to behind at N end DONE

7) Schedule replacement of 1/2" ball valves on ST rings. Ron G. knows which ones

8) Check out diff. plate MEL BUM ok

9) Tipping valve repair - BUM SAID HE WOULD FIX ok

10) Open Alk Duct and inspect from the grasshopper to the edge of the tower. (W/O) on metal cracks RANDY HAS JLG ON ITS WAY w/o WRITE. ABOUT 3" OR 4" IN BOTH. LEAN OUT

11) Keep close contact with brickers, brick location, inventory.

12) How much mud is left - R.D. ordered CAME IN - COAL SILO

13) Plastic wrap, box and mark the HZ 85 keys PUT IN DOL BOXES ! DOUG

14) GET COUNT ON OX + ACET. BOTTLES (2) PAT ORDERED

15) RETURN ACETONE DONE

16) DRYER POUR? HOW MUCH? THURSDAY

17) CALL WICKS DONE

18) START TAPPING ASHIMS BOXES -

TOTAL LBS 28,078

SOUTH ASSHOLE 5,063 LBS

OLD TIRE CHUTE 2,000 LBS

EAST WALL 10,518

Southdown
Lyons Cement Plant

BID DOCUMENT FOR FEBRUARY 1997 OUTAGE

OUTAGE IN APRIL 1997

Introduction

The enclosed bid package is for refractory and associated repairs at Southdown Inc. Lyons Colorado cement facility during the 1997 repair outage. Tentative schedule is for February-March time frame. Repairs may include but are not limited to: Kiln, Preheater Tower, Clinker Cooler, Spray Tower, and Dryer Combustion Chamber.

Firm bids must include the following projects:
1) Preheater Combustion Chamber Roof (Flash Furnace).
2) Kiln nose and brick removal.
3) Preheater Combustion Chamber Modification.
4) Kiln bullnose replacement (upper and lower).
5) Kiln feed table repair and installation of feed end castings.
6) IKN cooler refractory curbs removal and replacement.
7) Replace Cooler roof and walls.
8) Replace Dryer combustion chamber door and frame.
9) Alkali Bypass Take Off.
10) Assist with Kiln brick installation.

Some of the above projects could be changed to time and material, and some could be deleted depending upon refractory condition after final inspection. The majority will be completed. The following additional jobs may also be added on time and material basis. It can be assumed that the majority will be needed:

1) Tertiary Duct and take off.
2) Tertiary Damper replace metal and refractory.
3) Repair Dryer Combustion Chamber hair pin.
4) Repair Swirl Chamber refractory roof and floor as needed.
5) Replace doors and associated refractory on Preheater Tower.
6) Remove tire chute, replace metal and refractory.

It must be noted that the contractor is expected to complete both the firm bid and the time and material work in the fourteen work days available.

Each project listed for firm bids must have a separate bid total. Bids will be selected based upon the most competitive contract rates and time and material rates.

P. O. Box 529 • Lyons, Colorado 80540
(303) 823-2100 • Fax: (303) 823-2199

It is assumed that equipment such as tool trailer, company vehicles, etc that are needed for completion of firm bid jobs, will not be added in any way to the time and material jobs. It is also assumed that Southdown, Inc. will not be charged for higher skilled workers completing lower skilled jobs unless it is at our specific request. This is to promote having the proper number of each necessary skill on the job site for the amount of work forecast.

The successful bidder must supply Southdown, Inc. with a detailed comprehensive schedule including all firm bid projects based upon the fourteen work days available. The schedule needs to include the number of people per shift and how many shifts per project. This schedule will be coordinated with the facilities over all outage schedule. Bid expectation is such that the schedule must be met regardless of premium time, or the number of people that must be involved.

The contractor must follow all refractory manufacture and supplier recommended instructions concerning water percentage, mix time, temperature, etc. for placement of the refractory. If failure to follow these instructions causes valid concern on the refractory manufactures part, the entire job may be redone at no expense to Southdown, Inc.

The contractors project manager/superintendent must participate in a short scheduled meeting with plant personnel every morning and possibly every afternoon during the outage to discuss and coordinate the projects and schedules.

The contractor must comply with all Southdown, Inc., Lyons Plant, and MSHA safety rules and regulations which includes the proper MSHA training, and a short (30 min or less) site specific hazard recognition training meeting for each and every individual on the plant site. Additionally, each individual to be on the plant site during the outage must have in their possession at the beginning of the outage, a safety lock. A name tag will be provided by Southdown, Inc. to be signed by the employee and attached to the lock. Persons using someone else's lock, an unmarked lock, or no lock on any job that requires lockout/tagout may be dismissed from the plant site.

All of the above will be considered part of the firm bid.

All bids are due no later than 1 November 1996.

If you have any questions or would like to set up an appointment to visit this facility to go over this bid package please call Rich Adams or R.D. Smith at (303) 823-2128.

Rich: For 4-2-97

1) Clean up around the additive pit. Crushed gyp coming. OK

2) Need a W/O on the Gator. Done

3) Where are our two ladders? Found 1 ✓

4) Move anchors from the Storeroom to the Brickshed. Done ✓

5) Get a schedule from Bum for what gets poured when. ✓
   Al - Hole in cooler for eqpmt Wednesday -
   a) 1st CC pour Thursday
   b) 2nd CC pour Friday
   c) Dryer Thursday
   d) Cooler N and W walls Friday
   e) Cooler IKN Thursday
   f) Riser/tire chute — Sat
   g) Feed table - Sat
   h) Gun tail ring - Sun
   i) Repair damper/under damper
   j) Patch tertiary take off — Sat thru Sun

OK

6) Figure out brick alignment for brick return to the shed STARTed ✓

7) Face shields w ears

8) Check/order propane Done

9) Check N wall inside Riser @ Swirl ✓ 4" THick

10) Bolts for Tail castings - To short Done ✓

11) Move Kiln cast 26 To packhouse ASAP Done ✓

Double Burner gal eye, electric check out
Need Hose From Donkey To upper swirl - By
THURSDAY

FOR BUM
concrete bucket & forklift
morning at 10:00

WESTERN REFRACTORY CONSTRUCTION, INC.
4023 Mulligan Drive
Longmont, Colorado 80504

TIME SHEET

DATE 3/29/97 JOB # REPORT# 16

CUSTOMER Southwestern Cement

LOCATION Lyons Co

SCOPE OF WORK: Tire Chute 4

Employees's Name	Craft	st	1½	dt	tt	sub
Garson Curley	FM		12			1
Joe Garza	lbr		6			.5
Melvin Cheshire	lbr		6			.5
Time to remove lod & clean out & remove rubble from Tire chute & begin removal.						

EQUIPMENT	#	hrs	WRC	Rent
PICKUP				
TRUCK				
FLATBED TRAILER				
OFFICE TRAILER				
GUNITE MACHINE/100' Matl & Water hose				
MATERIAL HOSE - 50' section				
WATER HOSE - 50' section				
BULL HOSE - 50' section				
PREDAMPENER				
COMPRESSOR -825 1200				
WATER BOOSTER PUMP				
QUARTZ LIGHTS				
FORKLIFT				
CEMENT MIXER				
CONCRETE VIBRATOR				
JACK HAMMER	2	12	-	
BENCH RAMMER				
BRICKSAW				
BANDING MACHINE				
COMMUNICATION GEAR				
WELDING MACHINE	1	12	-	
CUTTING TORCH				
Brokk TEAROUT MACHINE				
HOBART MIXER				
ALL TERRAIN FORKLIFT				
SANDBLASTER, pot, hose & hood				
STUD GUN & CONTROLS				
WELDING LEAD				
MILEAGE: SEMI PICKUP ONE TON				
MATERIALS:				

RD Smith 3-31-97
Customer

Western Refractory

WESTERN REFRACTORY CONSTRUCTION, INC.
4023 Mulligan Drive
Longmont, Colorado 80504

TIME SHEET

DATE 3/31/97 JOB # _____ REPORT# 17

CUSTOMER Southwestern Cement

LOCATION Lyons Co.

SCOPE OF WORK:

Employees's Name	Craft	st	1½	dt	tt	sub
Kay Curley	B/MF	4	2			.5
Carlson Curley	BM	4	2			.5
Mike Rode	BM	4	2			.5
Joe Garza	Lbr	4	2			.5

Demo Tire Chute from feed end housing

EQUIPMENT	#	hrs	WRC	Rent
PICKUP				
TRUCK				
FLATBED TRAILER				
OFFICE TRAILER				
GUNITE MACHINE/100' Matl & Water hose				
MATERIAL HOSE - 50' section				
WATER HOSE - 50' section				
BULL HOSE - 50' section				
PREDAMPENER				
COMPRESSOR -825 1200				
WATER BOOSTER PUMP				
QUARTZ LIGHTS				
FORKLIFT				
CEMENT MIXER				
CONCRETE VIBRATOR				
JACK HAMMER	2	6	-	
BENCH RAMMER				
BRICKSAW				
BANDING MACHINE				
COMMUNICATION GEAR				
WELDING MACHINE	2	6	-	
CUTTING TORCH	2	6	-	
Brokk TEAROUT MACHINE				
HOBART MIXER				
ALL TERRAIN FORKLIFT				
SANDBLASTER, pot, hose & hood				
STUD GUN & CONTROLS				
WELDING LEAD				
MILEAGE: SEMI PICKUP ONE TON				
MATERIALS				

R D Smith 4-2-97
Customer

Western Refractory

TIME SHEET

WESTERN REFRACTORY CONSTRUCTION, INC.
4023 Mulligan Drive
Longmont, Colorado 80504

DATE 4/1/97 JOB # REPORT# 18

CUSTOMER Southwest Cement

LOCATION Lyons Co

SCOPE OF WORK: Riser Line Repairs

Employees's Name	Craft	st	1½	dt	tt	sub
Mike Rede	BM	2	1			
Derek Smith	lbr	8	4			1
Rodger Romero	lbr	8	4			1
East Wall / Tire Chute / N Wall						

EQUIPMENT	#	hrs	WRC	Rent
PICKUP				
TRUCK				
FLATBED TRAILER				
OFFICE TRAILER				
GUNITE MACHINE/100' Matl & Water hose				
MATERIAL HOSE - 50' section				
WATER HOSE - 50' section				
BULL HOSE - 50' section				
PREDAMPENER				
COMPRESSOR -825 1200				
WATER BOOSTER PUMP				
QUARTZ LIGHTS				
FORKLIFT				
CEMENT MIXER				
CONCRETE VIBRATOR				
JACK HAMMER	2	16	—	
BENCH RAMMER				
BRICKSAW				
BANDING MACHINE				
COMMUNICATION GEAR				
WELDING MACHINE				
CUTTING TORCH				
Brokk TEAROUT MACHINE				
HOBART MIXER				
ALL TERRAIN FORKLIFT				
SANDBLASTER, pot, hose & hood				
STUD GUN & CONTROLS				
WELDING LEAD				
MILEAGE: SEMI PICKUP ONE TON				
MATERIALS:				

ORD Smith 4-2-97
Customer

Western Refractory

WESTERN REFRACTORY CONSTRUCTION, INC.
4023 Mulligan Drive
Longmont, Colorado 80504

TIME SHEET

DATE 4/2/97 JOB # _____ REPORT# 20

CUSTOMER Southwest Cement

LOCATION Lyons Co.

SCOPE OF WORK: Demo in Riser.

Employees's Name	Craft	st	1½	dt	tt	sub
Derek Smith	labr	8	4			
Rodger Romero	labr	8	4			

Time to Demo on east wall of riser & around the skate also demo a small hole in cooler wall for camera port.

EQUIPMENT	#	hrs	WRC	Rent
PICKUP				
TRUCK				
FLATBED TRAILER				
OFFICE TRAILER				
GUNITE MACHINE/100' Matl & Water hose				
MATERIAL HOSE - 50' section				
WATER HOSE - 50' section				
BULL HOSE - 50' section				
PREDAMPENER				
COMPRESSOR -825 1200				
WATER BOOSTER PUMP				
QUARTZ LIGHTS				
FORKLIFT				
CEMENT MIXER				
CONCRETE VIBRATOR				
JACK HAMMER	2	16	-	
BENCH RAMMER				
BRICKSAW				
BANDING MACHINE				
COMMUNICATION GEAR				
WELDING MACHINE				
CUTTING TORCH				
Brokk TEAROUT MACHINE				
HOBART MIXER				
ALL TERRAIN FORKLIFT				
SANDBLASTER, pot, hose & hood				
STUD GUN & CONTROLS				
WELDING LEAD				
MILEAGE: SEMI PICKUP ONE TON				
MATERIALS:				

RD Smith 4-3-97
Customer

Western Refractory

Wed (Nights)

TIME SHEET

WESTERN REFRACTORY CONSTRUCTION, INC.
4023 Mulligan Drive
Longmont, Colorado 80504

DATE 4/2/97 JOB # R4297 REPORT#
CUSTOMER Southwest Portland Cement
LOCATION Lyons Co.
SCOPE OF WORK: Dryer

Employees's Name	Craft	st	1½	dt	tt	sub
Warren Jameson	Supt	3	1			.5
Guy Pelkey	GO	8	4			1
Ross Cervantez	BM	8	4			1
Jose Padilla	WRC Cmb	8	4			1
Finished steel work & anchors and set up for gunite						
Started setting up on riser where old Tire feed pipe was cut out.						

EQUIPMENT	#	hrs	WRC	Rent
PICKUP				
TRUCK				
FLATBED TRAILER				
OFFICE TRAILER				
GUNITE MACHINE/100' Matl & Water hose				
MATERIAL HOSE - 50' section				
WATER HOSE - 50' section				
BULL HOSE - 50' section				
PREDAMPENER				
COMPRESSOR -825 1200				
WATER BOOSTER PUMP				
QUARTZ LIGHTS				
FORKLIFT				
CEMENT MIXER				
CONCRETE VIBRATOR				
JACK HAMMER ~~& Zipper~~	~~1~~	~~12~~	~~—~~	
BENCH RAMMER				
~~BRICKSAW~~ Grinder	1	12	—	
BANDING MACHINE				
COMMUNICATION GEAR				
WELDING MACHINE	1	12	—	
CUTTING TORCH	1	12	—	
Brokk TEAROUT MACHINE				
HOBART MIXER				
ALL TERRAIN FORKLIFT				
SANDBLASTER, pot, hose & hood				
STUD GUN & CONTROLS				
WELDING LEAD	~~1~~			
MILEAGE: SEMI PICKUP ONE TON				
MATERIALS:				

R D Smith 4-4-97
Customer

Warren Jameson
Western Refractory

WESTERN REFRACTORY CONSTRUCTION, INC.
4023 Mulligan Drive
Longmont, Colorado 80504

TIME SHEET

DATE 4/4/97 JOB # REPORT# 19
CUSTOMER Southwest Cement
LOCATION Lyons Co
SCOPE OF WORK: Riser

Employees's Name	Craft	st	1½	dt	tt	sub
Pat Shannon	BM	8	0			
Gordon Cooley	BM	7	2			
Time to weld on the chute						

EQUIPMENT	#	hrs	WRC	Rent
PICKUP				
TRUCK				
FLATBED TRAILER				
OFFICE TRAILER				
GUNITE MACHINE/100' Matl & Water hose				
MATERIAL HOSE - 50' section				
WATER HOSE - 50' section				
BULL HOSE - 50' section				
PREDAMPENER				
COMPRESSOR -825 1200				
WATER BOOSTER PUMP				
QUARTZ LIGHTS				
FORKLIFT				
CEMENT MIXER				
CONCRETE VIBRATOR				
JACK HAMMER				
BENCH RAMMER				
BRICKSAW				
BANDING MACHINE				
COMMUNICATION GEAR				
WELDING MACHINE	2	4	—	
CUTTING TORCH	1	4	—	
Brokk TEAROUT MACHINE				
HOBART MIXER				
ALL TERRAIN FORKLIFT				
SANDBLASTER, pot, hose & hood				
STUD GUN & CONTROLS				
WELDING LEAD				
MILEAGE: SEMI PICKUP ONE TON				
MATERIALS:				

RD Smith 4-5-97
Customer

Western Refractory

TIME SHEET

WESTERN REFRACTORY CONSTRUCTION, INC.
4023 Mulligan Drive
Longmont, Colorado 80504

DATE 4/5/97 JOB # 134297 REPORT#
CUSTOMER South West Portland
LOCATION Lyons CO.
SCOPE OF WORK:

Employees's Name	Craft	st	1½	dt	tt	sub
Jose padilla	WRC Lab					
4/3/97 Dryer		8	4			1
4/4/97 Tire Chute		8	4			1

CHECK VS SHEET TIME SHEETS

EQUIPMENT	#	hrs	WRC	Rent
PICKUP				
TRUCK				
FLATBED TRAILER				
OFFICE TRAILER				
GUNITE MACHINE/100' Matl & Water hose				
MATERIAL HOSE - 50' section				
WATER HOSE - 50' section				
BULL HOSE - 50' section				
PREDAMPENER				
COMPRESSOR -825 1200				
WATER BOOSTER PUMP				
QUARTZ LIGHTS				
FORKLIFT				
CEMENT MIXER				
CONCRETE VIBRATOR				
JACK HAMMER				
BENCH RAMMER				
BRICKSAW				
BANDING MACHINE				
COMMUNICATION GEAR				
WELDING MACHINE				
CUTTING TORCH				
Brokk TEAROUT MACHINE				
HOBART MIXER				
ALL TERRAIN FORKLIFT				
SANDBLASTER, pot, hose & hood				
STUD GUN & CONTROLS				
WELDING LEAD				
MILEAGE: SEMI PICKUP ONE TON				
MATERIALS:				

RD Smith 4-6-97
Customer

Warren Kramer
Western Refractory

JOB PROGRESS REPORT

CUSTOMER Southwest Gmant Job #

ADDRESS Lyons Co Date 3/29/97

Report # 9

JOB PROGRESS Crew was all on we went to work shooting the area above the retrofit in the combustor once we had the scaffolding ready & completed all Ide vest & it looked good. also removed refractory from the line chute so it can be cut from the feed head & finaly got the new ass hole sleave installed & about 80% welded out. & cleaned up all rebound from top to bottom

DELAYS CAUSED BY OTHERS Kiln bricking is going real slow because of a shape brick + ect.

COMMENTS We got started putting prs together to build form in combustor.

WESTERN REFRACTORY CONSTRUCTION, INC.

Superintendent

JOB PROGRESS REPORT

CUSTOMER Southwest Cement Job #

ADDRESS Date 3/31/97

Report # 10

JOB PROGRESS Crew was in the unit to work in the cooler installing the last of the wall plates & casting behind them, also demoed the fine chute from feed end housing & then moved that crew over to the tertiary air duct to install new supports also formed in the combustor & had a good shift there also.

DELAYS CAUSED BY OTHERS

COMMENTS The bricking is still going slow but not from lack of effort or a lack of labor I think

WESTERN REFRACTORY CONSTRUCTION, INC.

Dan

Superintendent

JOB PROGRESS REPORT

CUSTOMER Southwest Cement Job #

ADDRESS Lyons Co Date 9/2/97

Report # 13

JOB PROGRESS CHW was all in we had a small safety meeting & went to work We continued forming on the additional area of the combustor & all but completed this also completed demo of the east wall of the risers & only lack a small amount on the area around the 2-line choke so that we can cut it again for an inland patch tertiary air duct & support ring is gone good & have new support welded out 100% of bent beam removed & reinstalled new one & only lack about 24 hrs to complete project

DELAYS CAUSED BY OTHERS

COMMENTS

WESTERN REFRACTORY CONSTRUCTION, INC.

Superintendent

JOB PROGRESS REPORT

CUSTOMER South west portland Job # 134297

ADDRESS Lyons CO. Date 4/5/97 Wight

Report #

JOB PROGRESS Riser Forming Welding Anchoes
Riser Steel Crack in Red paint

Forming Misc & welding anchors finished forming tail ring of feed shelf. & welded all anchors on bot ring

Also completed deck on area where slag chute was removed & also welded on misc. cracks on riser & mill valve

DELAYS CAUSED BY OTHERS

COMMENTS

WESTERN REFRACTORY CONSTRUCTION, INC.

Warren Jones

Superintendent

J.T. THORPE & SON, Inc.
Established 1906
ENGINEERS and CONTRACTORS

2000
OUTAGE

October 31, 1999

Southdown, Inc.
P.O. Box 529
5134 UTE Highway
Lyons, CO 80540

Attention: Royce (R.D.) Smith

Fax: (303)823-2199

Subject: Refractory Installation Proposal
January 2000 Outage
(JTT Q960)

Gentlemen:

We are pleased to furnish you with our quotation to install the refractory linings in your cement plant located in Lyons, Colorado as specified below:

Scope of Work:

Project #1 Kiln Nose and Brick Removal Kiln Nose)

- Use a Brokk Tearout Machine to remove 107' of kiln brick/coating and 5' of castable retainer ring
- JTT will mobilize two Brook machines on-site to perform this work. The second Brokk will be used as a stand-by machine
- Southdown, Inc. will be responsible for all muck out of the kiln brick/castable

Project #2 Assist with Kiln Brick Installation

- Provide two skilled brick masons for two 12 hour shifts per day for 4.5 days
- Provide two laborers for two 12 hour shifts per day for 4.5 days

Project #3 Kiln Feed Table Repair

- Remove the existing refractory form the bottom of the tail castings up the feed table to the vertical portion of the south side of the riser (maximum of 150 ft^2).
- Install new 8" 304 ss v-anchors
- Form and pump a 2800° F non-sticking castable refractory
- Remove the forms once the refractory has been allowed to cure

Project #4 IKN Cooler Refractory Curbs

- Remove the refractory curbs covering the IKN all the way to the steel on both sides of the cooler (maximum of 120 ft^2)
- Cover the new grates with ¾" plywood and a plastic sheet
- Install 8" 310 ss v-anchors, one per cover plate

572 West 800 South, Salt Lake City, UT 84101
801-359-9500 Fax: 801-363-6603

- Install two 2-½" PVC pipes for air blasters
- Install four cannon blast tube across the cooler
- Form and pump a coarse 3000° F low cement castable
- Remove the forms once the refractory has been allowed to cure

Project #5 Kiln and Cooler Bullnose Upper and Lower Roof

- Cover cooler grates with ¾" plywood and a plastic sheet
- Remove the refractory on the upper and lower bullnose, including all of the upper and portions of the lower roof section to the cooler walls (maximum of 270 ft^2)
- Remove all old refractory anchors
- Form and pump a 3000° F low cement castable
- Remove the forms once the refractory has been allowed to cure

Project #6 Replace Kiln Discharge Hood/Door Refractory

- Tearout all old refractory in the two hood doors
- Remove all old refractory anchors
- Install 7" 310 ss v-anchors @ 12" centers
- Form and pour a 3000° F low cement castable
- Remove the forms once the refractory has been allowed to cure (Not needed)

Project #7 Replace Metal on Top of the Tertiary Duct from the Take Off to the Drop Out

- Install a bulk head in the end of the tertiary take off
- Remove 29' of the upper half of the tertiary air duct refractory
- Remove and replace the steel in 29' of the upper half of the tertiary air duct
- Form and pump a 2600° F low cement castable
- Remove the forms once the refractory has been allowed to cure

Additional Projects

The following are the anticipated additional projects to be completed during the shutdown. These projects will be completed on a time and materials basis per the rates submitted with this proposal.

- Tertiary duct repair including expansion joints
- Tertiary damper metal and refractory
- Repair SS ring at alkali quench chamber
- Riser duct refractory repair as needed
- Install new chains and steel plate at drop out discharge
- Repair tertiary takeoff from the cooler as needed
- Dryer feed pipe and tail ring, repair refractory
- Install tire feed chute

Pricing:

J.T. Thorpe & Son, Inc. will furnish all the necessary Labor and Equipment required to install the above refractory for the following price:

ONE HUNDRED FIFTY TWO THOUSAND NINE HUNDRED THIRTY SIX DOLLARS ($152,936)

To; John Lohr
From: R.D. Smith/Rich Adams
Subject: 2000 Kiln Outage Work Scope
Date: 09-01-99

PROJECT SCOPE FOR 2000 KILN OUTAGE

~~Relocate burner floor crane controls prior to outage~~

The following list is the anticipated work scope for the January 2000 kiln outage. This list includes all known projects as of this date.

Firm bids will be required on the following projects:
1) Kiln nose and 115' of brick removal.
2) Assist with Kiln brick and castable retainer installation.
3) Kiln and cooler bullnose including upper and lower roof.
4) Remove and replace IKN refractory, install blasters.
5) Kiln feed table
6) Replace metal and refractory on the top of the tertiary duct from the take off to the dropout.
7) Modify the tertiary take off from the cooler.
8) Modify the bottom of the dropout chamber, and the pipe to the cooler.
9) Install whole tire feed chute at the kiln feed hood. (move To T+M)

Some of the above projects could be changed to time and material, some could be deleted depending upon refractory condition after final inspection. The majority will be completed.

The following jobs may also be added on time and material basis. It can be assumed that the majority will be needed:

1) Tertiary Duct repair including expansion joints.
2) Tertiary Damper replace metal and refractory on the lower section.
3) Replace diffuser plate.
4) Repair dryer combustion chamber ground level door area.
5) Dryer feed pipe and tail ring refractory.
6) Repair riser duct refractory as needed.
7) Repair/replace alkali take off metal/refractory.
8) Repair kiln discharge hood/doors.

R.D. Smith
Production Manager

Rich Adams
Production Team Leader

Introduction

The enclosed bid package is for refractory and associated repairs at Southdown Inc. Lyons Colorado cement facility during the January 2000 repair outage. Tentative schedule is 14 days. feed to feed. Work includes but is not limited to: Kiln, Preheater Tower, Clinker Cooler, Spray Tower, and Dryer Combustion Chamber.

Firm bids must include the following projects:

1) Kiln nose and brick removal.
2) Assist with kiln brick installation.
3) Kiln feed table repair.
4) IKN cooler refractory curbs and installation of blow pipes.
5) Kiln and cooler bullnose including upper and lower roof
6) Replace kiln discharge door refractory
7) Replace 29' of metal and refractory on top of the tertiary duct.

Some of the above projects could be changed to time and material, some could be deleted depending upon refractory condition after final inspection. The majority will be completed.

The following additional projects may also be added on time and material basis. It can be assumed that the majority will be needed:

1) Tertiary duct repair including expansion joints.
2) Tertiary damper metal and refractory.
3) Repair SS ring at alkali quench chamber
4) Riser duct refractory repair as needed.
5) Install new chains and steel plate at drop out discharge.
6) Repair tertiary takeoff from the cooler as needed.
7) Dryer feed pipe and tail ring, repair refractory.
8) Install tire feed chute.

It must be noted that the contractor is expected to complete both the firm bid and the T&M work in 13 work days. On time or early completion will result in a bonus of 5% of the total final agreed upon invoice amount.

Each project listed for firm bids must have a separate bid total. Bids will be selected based upon the most competitive contract and T&M rates.

T and M projects

1) Tertiary duct repair including expansion joints.
   A) Some welding at the joints may be required.
   B) K-wool in the expansion joints may need to be replaced.

2) Tertiary damper metal and refractory repair.
   A) The bottom 12" section metal will need to be repaired.
   B) AA 22 or similar ram will be needed.

3) Repair SS ring at alkali quench chamber: 12 man hrs welding.

4) Riser duct refractory repair as needed: Bulkheads will be required to prevent material from falling into the kiln.

5) Install new chains and stainless steal plate at drop out discharge.

6) Repair tertiary take off from the cooler as needed.
   A) Estimated metal work 1 man 3 days
   B) Patch refractory

7) Dryer feed pipe and tail ring, repair refractory
   A) Remove old refractory, anchors, and debris
   B) Weld new anchors, and gun material

8) Install tire feed chute to kiln feed hood
   A) Remove metal and castable (30"x 4')
   B) Install new prefabricated frame and flange

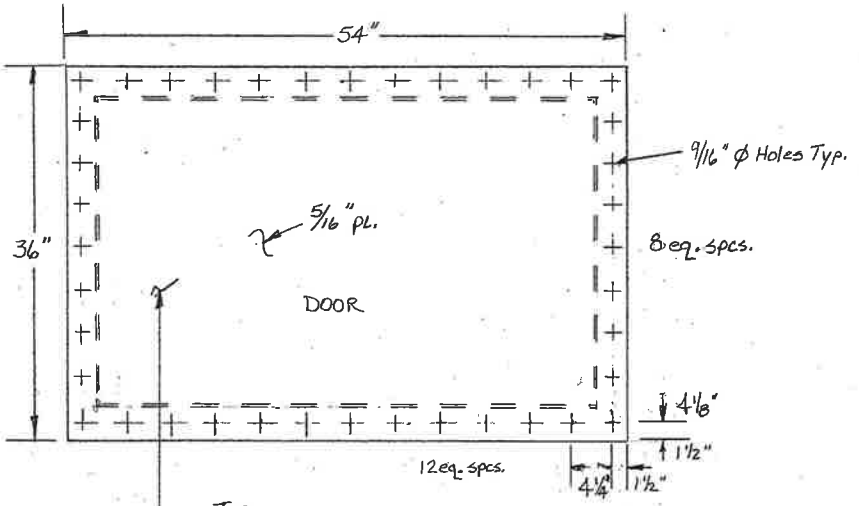
9) Dryer combustion chamber lower south wall
   A) Remove old refractory and anchors
   B) Weld new anchors, form and pour

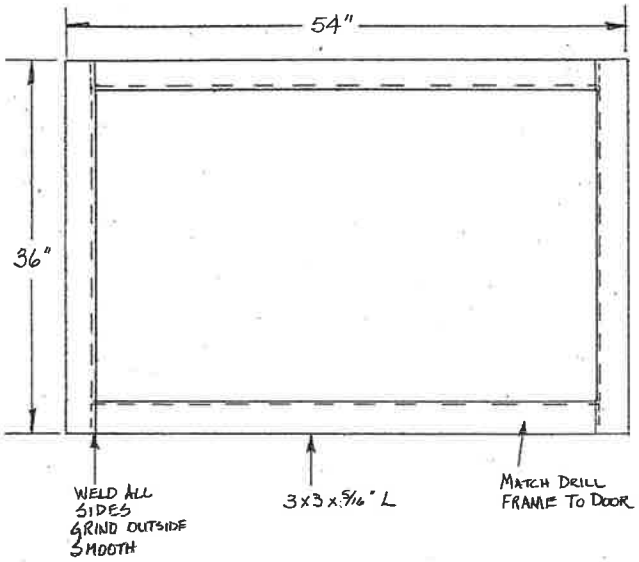
10) Patch/repair tertiary duct refractory at 95'
   A) Scaffolding will be needed in the drop out for access

11) Repair refractory inside the heat exchanger duct

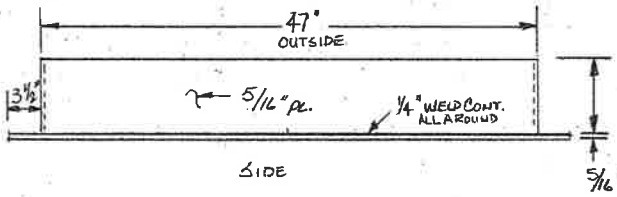
12) Repair metal at tertiary damper housing

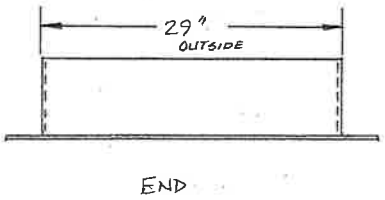
Most jobs will be determined at the time of the outage.





4.25





3'x4' = 12 SQ FT x .58 = 7 CU FT

7 CU FT x 160 LBS/CU FT = 1120 LBS

(7" = .58)

NO.	DATE	REVISION	BY	CH'D	MF'D	W.O.NO.

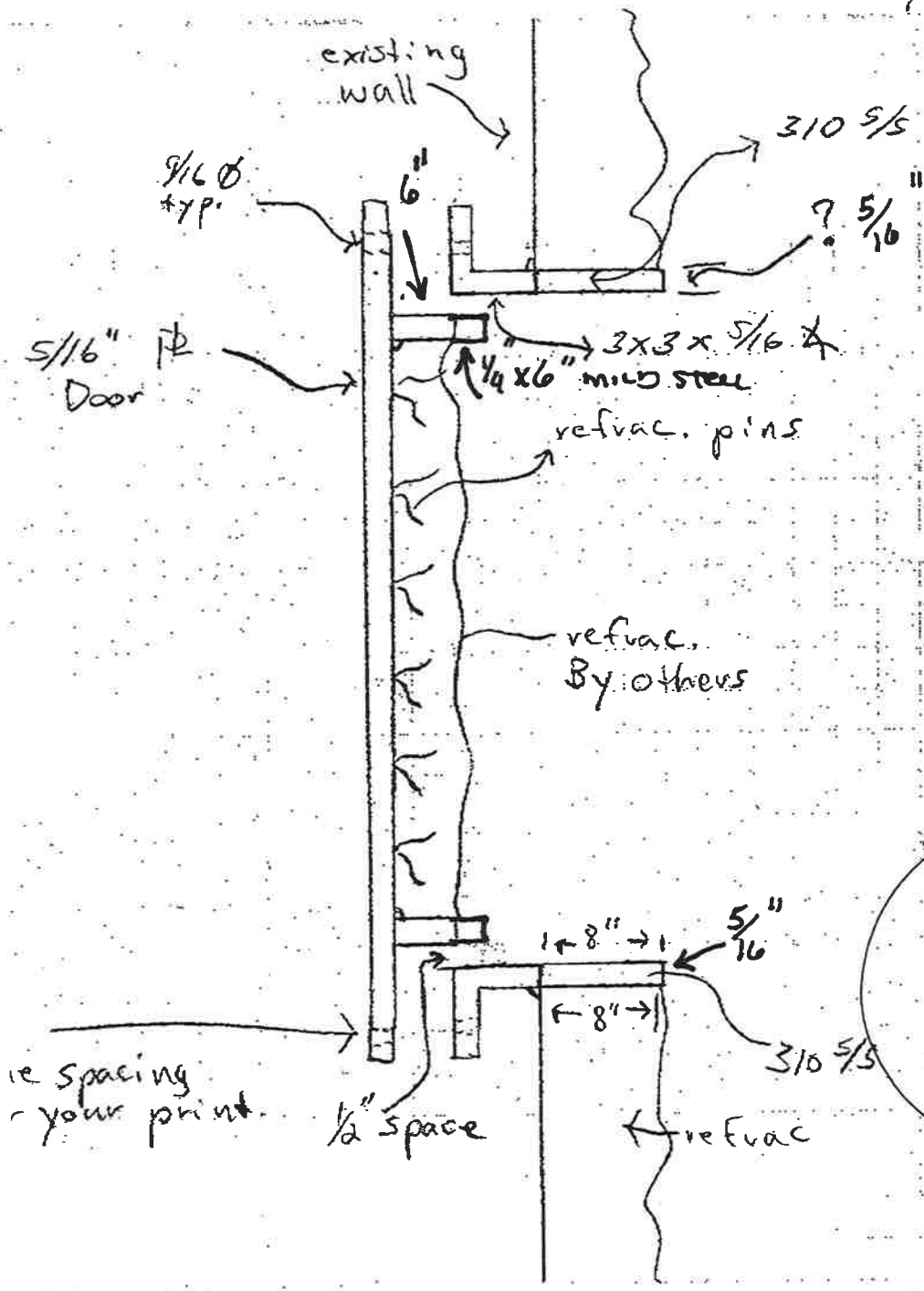
SCALE 3/4" = 1'		SOUTHWESTERN PORTLAND CEMENT COMPANY	DEPARTMENT Tire Feed Chute Door
DESIGN R. Adams			
DRAWN Sept. 23, 1999	WORK ORDER NO.		
CHECKED			
REVIEWED		LOS ANGELES CALIFORNIA	DRAWING NUMBER / REV
MICROFILMED DATE	PLANT		

Fax 303-823-2100

ATTN.

R.D.

Dave Kretesch



R.D. I Belive This is what you want. If it is sign and fax back so I can yet you numbers.

OK
10-13-99

ie spacing
your print.

STOCK NO.	DESCRIPTION	DATE	EQUIPMENT #	QTY.	UNIT OF MEASURE
470300	1/8" 7018 welding Rod	1-5-00	545-1	EA	50 LBS

NOTES:

Qty. left on hand 4

Reorder ☐

COMMENTS:

MATERIAL REQUISITION

Tire Feed chute

DELIVERED BY

STOCK CLERK

RECEIVED BY

R Adams

TO BE USED FOR

☐ PREVENTATIVE MAINTENANCE

☐ EMERGENCY BREAKDOWN REPAIR

☐ OTHER _____

STOCK NO.	DESCRIPTION	DATE	EQUIPMENT #	QTY.	UNIT OF MEASURE
470440	1/8" STAINTRod ID. welding Rod	1-5-00	545-1	EA	10 LBS

NOTES:

Qty. left on hand 12

Reorder ☐

COMMENTS:

MATERIAL REQUISITION

Tire Feed chute

[redacted]

DELIVERED BY

STOCK CLERK

RECEIVED BY

R Adams

TO BE USED FOR

☐ PREVENTATIVE MAINTENANCE

☐ EMERGENCY BREAKDOWN REPAIR

☐ OTHER _____

STOCK NO.	DESCRIPTION	DATE	EQUIPMENT #	QTY.	UNIT OF MEASURE
170150	1/4 Air	1/4/00	586-7	50	ea

NOTES:

left on hand _____

rder ☐

COMMENTS:

MATERIAL REQUISITION

Tire Feed chute

DELIVERED BY

STOCK CLERK

RECEIVED BY

ROS JAN 4 REC'D

TO BE USED FOR

☐ PREVENTATIVE MAINTENANCE

☐ EMERGENCY BREAKDOWN REPAIR

☐ OTHER _____

To: R. D. Smith
From: R. Adams
Subject: Castable needed for 2000 outage
Date: 10-22-99

1) Kiln bull nose including upper and lower roof 3000 degree material 43,824 lbs + 3000 lbs of insulating castable 2" thick
2) Kiln hood doors 3000 degree material 14,375 lbs
3) Tertiary duct from cooler 3000 degree material 10,890 lbs
4) IKN curbs 3000 degree material 25,600 lbs
5) Dryer north wall 3000 degree material 15,360 lbs

For a total of 110,049 lbs of 3000 degree material

6) Kiln feed table 2800 degree (zirconium) material 24,420 lbs
7) Tire feed chute 2800 degree material 1120 lbs
8) Tertiary duct 2800 degree material 36,300 lbs
9) Kiln retainer ring 2800 degree material 16,000 lbs

For a total of 77,840 lbs of 2800 degree material

10) Riser patch work 13' 2600 degree material 24,960 lbs total

WESTERN REFRACTORY CONSTRUCTION, INC.
4023 Mulligan Drive
Longmont, Colorado 80504

TIME SHEET

DATE 1/4/00 JOB # 4-8-2001 REPORT# Riser Tire Schedule

CUSTOMER Southdown

LOCATION Lyons Colorado.

SCOPE OF WORK: Install tire Schedule

Employees's Name	Craft	st	1½	dt	tt	sub
Thomas Reel	Wld	8	4		10	2
Jones Lee	Wld	8	4		10	2
Raymond Chee	Lbr	8	4			1
Daniel Charlie	Lbr	8	4			1

EQUIPMENT	#	hrs	WRC	Rent
PICKUP				
TRUCK				
FLATBED TRAILER				
OFFICE TRAILER				
GUNITE MACHINE/100' Matl & Water hose				
MATERIAL HOSE - 50' section				
WATER HOSE - 50' section				
BULL HOSE - 50' section				
PREDAMPENER				
COMPRESSOR -825 1200				
WATER BOOSTER PUMP				
QUARTZ LIGHTS				
FORKLIFT				
CEMENT MIXER				
CONCRETE VIBRATOR				
JACK HAMMER	2	12	-	
BENCH RAMMER				
BRICKSAW				
BANDING MACHINE				
COMMUNICATION GEAR				
WELDING MACHINE	1	12	-	
CUTTING TORCH	1	12	-	
Brokk TEAROUT MACHINE				
HOBART MIXER				
ALL TERRAIN FORKLIFT				
SANDBLASTER, pot, hose & hood				
STUD GUN & CONTROLS				
WELDING LEAD				
MILEAGE: SEMI PICKUP ONE TON				
MATERIALS:				

RD Smith 01-05-00
Customer

Western Refractory

WESTERN REFRACTORY CONSTRUCTION, INC.
4023 Mulligan Drive
Longmont, Colorado 80504

TIME SHEET

DATE 1/4/00 JOB #48-2001 REPORT# 1-Tire Schedule

CUSTOMER Seattlean

LOCATION Install new Tire Schedule

SCOPE OF WORK:

Employees's Name	Craft	st	1½	dt	tt	sub
Gary Harrison	BMF	3	2			
Edison Harvey	WH	3	2			
Max McKee	Ibr	3	2			

EQUIPMENT	#	hrs	WRC	Rent
PICKUP				
TRUCK				
FLATBED TRAILER				
OFFICE TRAILER				
GUNITE MACHINE/100' Matl & Water hose				
MATERIAL HOSE - 50' section				
WATER HOSE - 50' section				
BULL HOSE - 50' section				
PREDAMPENER				
COMPRESSOR -825 1200				
WATER BOOSTER PUMP				
QUARTZ LIGHTS				
FORKLIFT				
CEMENT MIXER				
CONCRETE VIBRATOR				
JACK HAMMER				
BENCH RAMMER				
BRICKSAW				
BANDING MACHINE				
COMMUNICATION GEAR				
WELDING MACHINE	1	4	-	
CUTTING TORCH	1	4	-	
Brokk TEAROUT MACHINE				
HOBART MIXER				
ALL TERRAIN FORKLIFT				
SANDBLASTER, pot, hose & hood				
STUD GUN & CONTROLS				
WELDING LEAD				
MILEAGE: SEMI PICKUP ONE TON				
MATERIALS:				

RD Smith 01-05-00
Customer

Western Refractory

WESTERN REFRACTORY CONSTRUCTION, INC.
4023 Mulligan Drive
Longmont, Colorado 80504

TIME SHEET

DATE 1/11/2000 JOB # 49-2001 REPORT# Time Sheet 2000

CUSTOMER Southdown

LOCATION Lyons Co.

SCOPE OF WORK:

Employees's Name	Craft	st	1½	dt	tt	sub
James Jordan	lbrf	4	2			4
Jackie Beverly	lbr	4	2			4
Randall Escotto	lbr	4	2			
Melvin Charles	lbr	4	2			
* Time to cast door & set up						
Move equipment in place						

EQUIPMENT	#	hrs	WRC	Rent
PICKUP				
TRUCK				
FLATBED TRAILER				
OFFICE TRAILER				
GUNITE MACHINE/100' Matl & Water hose				
MATERIAL HOSE - 50' section				
WATER HOSE - 50' section				
BULL HOSE - 50' section				
PREDAMPENER				
COMPRESSOR -825 1200				
WATER BOOSTER PUMP				
QUARTZ LIGHTS				
FORKLIFT				
CEMENT MIXER	1	2	—	
CONCRETE VIBRATOR	1	2	—	
JACK HAMMER				
BENCH RAMMER				
BRICKSAW				
BANDING MACHINE				
COMMUNICATION GEAR				
WELDING MACHINE				
CUTTING TORCH				
Brokk TEAROUT MACHINE				
HOBART MIXER				
ALL TERRAIN FORKLIFT				
SANDBLASTER, pot, hose & hood				
STUD GUN & CONTROLS				
WELDING LEAD				
MILEAGE: SEMI PICKUP ONE TON				
MATERIALS:				

RD Smith 01-12-00
Customer

Western Refractory